



Fighting to the beat: Multi-instrumental adaptive background music approach[☆]

Ibrahim Khan^a,^{ID},^{*} Thai Van Nguyen^a,^{ID}, Ruck Thawonmas^b,^{ID}

^a Graduate School of Information Science and Engineering, Ritsumeikan University, Ibaraki, Japan

^b College of Information Science and Engineering, Ritsumeikan University, Ibaraki, Japan

ARTICLE INFO

Keywords:

Adaptive BGM
Rule-based adaptive background music
Background music
Music modification
Instrument modulation

ABSTRACT

This paper presents our approach to enhance the background music (BGM) in DareFightingICE, a fighting game research platform, by adding a rule-based adaptive BGM. The adaptive BGM consists of three different groups of instruments playing the BGM provided in the DareFightingICE platform. The BGM adapts by changing the volume of the groups of instruments. Each group is connected to a different element of the game. We then run experiments to evaluate the adaptive BGM by using a deep reinforcement learning AI that only uses audio as input (Blind DL AI). The results show that the performance of Blind DL AI improves while playing with the adaptive BGM as compared to playing with a baseline non-adaptive BGM. We also conduct two user studies, a perception study, and a playing study. The results from these studies show that not only human players can recognize the adaptation of the BGM, they also perform better with adaptive BGM as compared to non-adaptive BGM. Lastly, the adaptive BGM scores higher than the non-adaptive on audio aesthetics and enjoyment factors making it overall a better BGM.

1. Introduction

Video games are becoming more and more popular, with this growth being quick and constant. Artificial intelligence (AI) has been linked to video games for some time now [1]. The study of AI in video games has contributed to numerous advances in AI [2]. Video game designers are working to improve their games as the industry's popularity soars, aiming to provide players with a favorable experience. These enhancements can take many different forms, such as adjustments to the gameplay, games with appealing visuals, or games with excellent music or sound effects. The ultimate objective, regardless of the kind of enhancement, is to improve the players' overall experience.

This study focuses on using music in video games to improve player experience by giving players more information about the game. An immersive gaming experience is aided by background music (BGM) [3] or music [4]. The target genre in this research is fighting games. In fighting games, players engage in one-on-one battles, utilizing various attacks and skills to defeat their opponent, whether it is another player or a computer-controlled character. Traditionally, fighting games are two-dimensional or two-and-a-half-dimensional, which means that the players can only move in two dimensions (left or right). In fighting

games, BGM should be able to change in response to game situations, and doing this creates suspense and excitement as the players engage in combat. Commonly, a fighting game's BGM is implemented using pre-composed songs that loop, which often falls short of accurately capturing the intensity of combat. A few examples of commercial fighting games are Street Fighters [5], Tekken [6], and Mortal Kombat [7].

This research uses the DareFightingICE platform, which is a fighting game platform [8]. It is an audio-enhanced version of the FightingICE platform [9] and was used in the DareFightingICE Competition, which had two tracks: the Sound Design Track and the AI Track. The competition had been run at the IEEE Conference on Games from 2022 to 2024. More details about DareFightingICE will be given in a later section.

Information below is given to better understand what is an adaptive BGM in this research.

Adaptive Music reacts dynamically to in-game events and player actions, adjusting elements such as instrumental layers, tempo, and effects to match the intensity or atmosphere of the gameplay. In this research, the proposed adaptive music can be described as a system that allows the music's loudness to be

[☆] The authors of this paper would like to express their gratitude to Chuang Boyu, Shouchen Ye, Tian Yuchen, and Jun Tanabe from the Intelligent Computer Entertainment laboratory, Ritsumeikan University for their help in translating the user study details into Japanese for Japanese-speaking participants. This work was supported in part by Grant-in-Aid for Scientific Research (C), No. 25K15412.

^{*} Correspondence to: Ritsumeikan University, Graduate School of Information Science and Engineering, Intelligent Computer Entertainment Laboratory, Japan.
E-mail addresses: gr0556vx@ed.ritsumei.ac.jp (I. Khan), gr0557fv@ed.ritsumei.ac.jp (T. Van Nguyen), ruck@is.ritsumei.ac.jp (R. Thawonmas).

altered in real-time based on control inputs, such as player health and the distance between players.

Static Music on the other hand, consists of pre-composed tracks that play consistently without changing in response to gameplay. In this research, static music refers to music whose loudness remains constant and is unaffected by external factors, such as the player's health.

The goal of this research is to create an adaptive BGM that gives players information about the state of the game. It is done by modifying the BGM of DareFightingICE. This research is an extension of a previous study [10], focusing on giving players with and without vision information about the state of the game through the context-aware BGM. We picked fighting games as it has been shown to be popular among VIPs [11]. This research is the first one to use multiple instruments in the adaptation of BGM. The contributions of our work are as follows:

1. This is the first study to focus on adaptive music in fighting games and it is also the first to use multiple instruments in the adaptation of BGM.
2. Our approach demonstrates that adaptive BGM enhances the overall enjoyment of the game compared to static BGM.
3. Our adaptive BGM is not only more informative but also more aesthetic than its non-adaptive counterpart.

2. Related work

Since our research touches multiple topics, we divide this section into four parts.

2.1. Sound in video games

Research has been done on the importance of sound in video games [12,13]. It has been established through research that music in video games has an impact on a person's daily life, personality, and such [14]. The reason for this impact of music on a person can be due to a person's attachment to the music used or played in a game. Sound in video games in terms of voiceover (audio dialogue by the player or the non-playable characters) is known to be more engaging for the players as compared to games without them [15]. These voiceovers also help in reminding the player of the information provided in the game. The BGM in the game also has an important role to play as well. It is observed that the immersion of the players with low performance improve when the game has BGM as compared to the game without it [16].

Similarly, it is known that the sound effects in video games are the most important element to be able to engage the players [17]. A simple bad example of how not to match the tone of the game would be to play calm music when the game is at a high-tension point. Going from general music to specific sound effects and audio cues, it has been observed that when there are directional or 3D sound effects in a game environment it helps the players to know more about their environment, such as having an idea of the location of other players [18]. These 3D sound effects include sounds like footsteps and other movement sounds.

Till now this sub-section has been about sound in video games. Now moving from sound in video games to sound designs, because a sound design is a set of all the sound effects and music in the game as well as the logic on when and how these sound effects and music will be played. The research when it comes to sound designs in video games has been very little. Even fewer have had good results that can be useful in the creation of a sound design. Scarce as the research may be, sound designs have been proposed for game genres such as first person shooter and real-time strategy [19].

2.2. Games for visually impaired players and fighting games

In this sub-section we talk about research related to visually impaired players (VIPs) and fighting games that have support for such

players. The reason to look at fighting games that have support for VIPs is because this research focuses on fighting games. Another reason to focus on fighting games when it comes to VIPs is that a previous work observed that fighting games are among the second most played games in VIPs [11]. Some research on serious games (a game with a purpose like an educational game) has been done in the past involving VIPs. Previous work shows that audio cues in a serious game are able to convey spatial information about a room to such players [20]. These audio cues are played as VIPs explore a room in the game. Another work introduces NavStick, an audio-based tool for looking around within virtual environments to help VIPs [21].

Moving from research on games for VIPs to fighting games with VIPs compatibility. The reason to focus on commercial fighting games is that there has been little to no research in fighting games when it comes to VIPs to the best of our knowledge. We concentrate on three fighting games here that have such compatibility to the best of our knowledge. Tekken 7 [22] is the first, and least compatible. The distinct sounds that each character in this game makes it easy to identify and recall. The fighting game we include in the Mortal Kombat fighting game series is Mortal Kombat 11 [23]. This game is much more beneficial since, in addition to having distinct sounds for each character, the character selection screen provides distinct audio cues for each character. The fighting game Killer Instinct [24] features stereo sound and several sound effects, like a sound effect that plays to assist VIPs at the end of a stage. The fourth and final game is Street Fighter 6 [25], the latest game in the Street Fighter series. This game has many different audio cues to help VIPs like audio cues for the distance between the characters and different audio for different levels of hits.

VIPs can and have been able to play all of the aforementioned commercial games. However, there is still a lot of room for improvement. Since these games were not created with VIPs in mind, or at least their publishers did not say as much, they are not particularly user-friendly for VIPs.

2.3. Adaptive audio in video games

Research on adaptive music in video games has been done in the past. A paper explored various methods for creating real-time adaptive music in interactive environments, focusing on the musical aspects and their impact on player experience [26]. Another study investigated the use of adaptive music to reflect the emotional states of game characters, enhancing player immersion and emotional engagement [27]. Another piece of research claimed that their sonification method could improve player engagement by adapting music to each game situation; this audio system that adapts environmental music to emotions associated with the ongoing narrative [28]. A study created adaptive music, keeping players' emotions in mind while creating it, which was shown to be better than the original music in the game [29]. Another study found that players tend to enjoy the game more with adaptive music as compared to static music [30]. Lastly, another piece of work used the player's actions and the events in the game to create adaptive music by changing the tempo of the music and found that by doing so the playing experience was enhanced [31].

The above research has shown the importance of both music and adaptive music in video games. However, there has been little to no research when it comes to adaptive music in fighting games.

2.4. Adaptive audio in fighting games

As mention before there has been little to no research on adaptive music in fighting games. Nevertheless, there are fighting games that have some elements of adaptive music. Adaptive or dynamic music, as it is called in commercial fighting games, has been used in a few fighting games to the best of our knowledge. The best examples are Killer Instinct [24], Tekken 7 [22], and Them's Fightin' Herds [32]. For Tekken 7, many arenas where players fight have a more intense BGM

that plays for the last round. Them's Fightin' Herds adds in the winning character's theme in the BGM that is being played. Lastly, Killer Instinct has multiple layers to its adaptive BGM that change according to the movement of the players as well as the actions of the players.

2.5. DareFightingICE

DareFightingICE [8] was designed to open new doors to developers and researchers as it was the first platform that focuses on VIPs as well as AI agents that only use audio as input (Blind AI agents). The important features of this platform are given below.

2.5.1. Sound design

DareFightingICE has a sound design that is made with VIPs in mind. It has 52 sound effects including the BGM and has directional audio to play these sound effects in the direction and location of the characters' fight. In addition, the sound design has three special sound effects (SSEs) designed especially for VIPs. The first SSE is a border alert sound that plays when a character touches the border of the stage; it is played on the left if the character touches the left border and vice versa. The others are an energy increase sound that plays every time a character's energy is increased by 50 and a heartbeat sound that plays when the health points (HP) of a character is below 50; both sounds are played on the left for player 1 and on the right for player 2.

2.5.2. AI interface

One notable feature of DareFightingICE is its capability to implement an AI agent and deploy it as a fighting agent against its opponent, either an AI agent or a player character. The platform initially supported AI agent implementation using Java programming by compiling the AI agent code and integrating it into the platform as a plugin, which is then included upon the initial loading of the game. Later, the platform facilitated AI agent implementation using Python programming through Py4J integration. However, there was been a notable issue regarding high latency on Linux between the Py4J client (the AI agent) and the Py4J server (the platform). This problem was addressed through the use of gRPC Remote Procedure Call (gRPC), resulting in a latency reduction of up to 65% for Python-based AI agents, alongside improved stability [33].

3. Rule-based adaptive BGM

As mentioned before, this research is a continuation of a previous work [10] where the authors proposed an adaptive BGM,¹ that adapts to the consequences of both players' actions. The proposed adaptive BGM consists of three different groups of instruments playing the BGM from the winning sound design of the 2022 DareFightingICE Competition alongside the rest of the sound effects in the sound design. The three different groups of instruments separated from the BGM of the winning sound design are the drums, strings, and others; the others mainly consist of electric guitars with added bass.

The three different categories of instruments that are separated from the BGM are the drums, the strings, and the others which mainly consist of electric guitar with added bass. These groups of instruments are separated using an AI tool called Moise [34]. This tool is selected since it is most suited to our needs. Fig. 1 shows Moise in action. The BGM adjusts by altering the loudness of the groups of instruments. This adjustment happens when the game elements connected to these instrument groups change. The elements in use are both players' health points (HP) and the distance between the two players (PD), where the HP is the number of hits a player can take before losing. The design of the proposed adaptive BGM is illustrated in Fig. 2.

¹ Code of the adaptive BGM added in DareFightingICE 6.2 version: <https://tinyurl.com/Code-A-BGM>.

While various techniques, such as tempo adjustment, have been effective in aligning music with in-game events and enhancing player engagement [31], volume or loudness modulation offers a unique approach in a fighting game setting, where immediacy and intensity are crucial. Unlike tempo or pitch changes, which can significantly affect the mood or style of the music — especially in the case of pitch — [35] adjusting volume allows for an immediate response to game dynamics, such as signaling high-stakes moments, without altering the overall aesthetic or making substantial changes to the harmony or mood. Loudness adjustments also require less technical knowledge of music theory, making our approach more accessible to a wider range of game designers.

In Fig. 2, three different groups of instruments play together to compose the BGM, and every instrument group is connected to a different game element. The drums, the strings, and the others are connected to the PD, HP of Player 1, and HP of Player 2, respectively. Note that in our setting, the human player plays Player 1, while the AI plays Player 2. These instruments are connected to their respective game elements from most to least important. Because the drums are different from the strings and others and are thus easiest to distinguish, this group is connected to the PD. This is also based on a previous study showing VIPs rank position and orientation the highest among all aspects of spatial awareness in video games [36]. Next, the strings are connected to the HP of Player 1 because, through our empirical testing, they are more distinguishable than the others, and it is more important to know information about the player's own HP than the opponent's.

In Fig. 3, the volume of the drums is at its maximum when the PD is close to zero, meaning the players are very close to each other, and it becomes lower the further. In addition, the groups of instruments connected to the HP of both players have their volume turned up as the HP gets lower and will be at maximum volume when the HP is minimum. The proposed adaptive BGM is designed in this fashion to give helpful information to both human and audio-input AI players.

The maximum HP for each player per round is 400, and the maximum PD after which volume is not changed is 600 pixels horizontally with the maximum PD being 800. The HP of both players and their PD is divided into different levels to decrease or increase the volume of the respective instruments gradually. The levels of not only instruments but also players' HP, and PD are empirically selected due to the lack of existing research when it comes to adaptive BGM using volume modulation.

The drums' volume levels are around 85%, 75%, 60%, 50%, 40%, 30%, and 10%, linked to [0, 30], (30, 60], (60, 300], (300, 400], (400, 500], (500, 600], (600, 800] PD intervals, respectively; and the volume levels of both strings and others are approximately set to 85%, 70%, 65%, 50%, 45%, 35%, and 10%, linked to HP intervals of [0, 50], (50, 100], (100, 150], (150, 200], (200, 250], (250, 300], and (300, 400], respectively.

4. Objective evaluation

For the evaluation of the adaptive BGM, we did an objective evaluation, the details of which are the following.

We used a deep reinforcement learning blind AI agent (Blind DL AI) [37] for the objective evaluation. We trained Blind DL AI with and without the adaptive BGM as part of our evaluation process, and then we compared their performances. We hypothesized that the adaptive BGM will provide valuable information if Blind DL AI performs better while using it. Three sound designs — the default, runner-up, and winning sound designs from the 2022 DareFightingICE Competition — were used in the experiments.

4.1. Blind DL AI

Blind DL AI is a deep reinforcement learning AI agent that only uses audio as input. Since it only uses audio as input, it is used in this

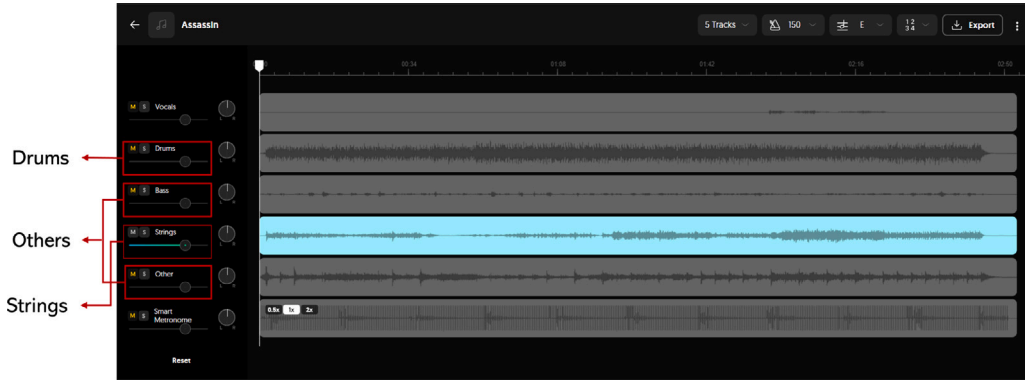


Fig. 1. A screenshot of Moise [34] being used to separate the BGM.

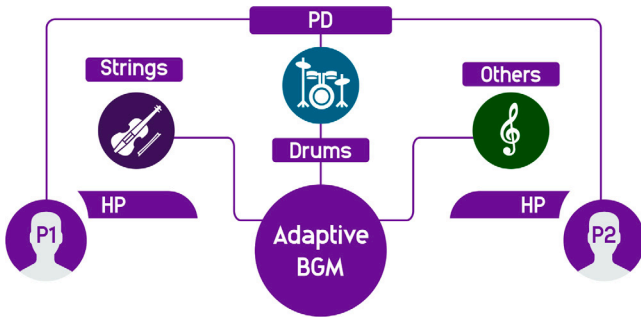


Fig. 2. Design of adaptive BGM.

```

For Drums:
    if Pos.diff decreases
        Drums.volume.increaselevel
    Else
        Drums.volume.decreaselevel
For Strings:
    if P1.HP decreases
        Strings.volume.increaselevel
For Others:
    if P2.HP decreases
        Others.volume.increaselevel

```

Fig. 3. Adaptive BGM rules.

research in the evaluation part of the adaptive BGM. For the sake of a self-contained presentation, an overview of Blind DL AI is given below summarized from previous work [37].

Blind DL AI has three different ways to handle audio data provided by DareFightingICE. These different ways are three audio encoders called, one-dimensional Convolutional neural network (1D-CNN), Fast Fourier Transformation (FFT), and Mel Spectrogram (Mel-Spec). Their architectures are shown in Fig. 4.

1D-CNN: Every eighth sample is used to downsample the input audio data, which is then put into two 1D convolution layers. The difficulty of the computations is decreased by downsampling. A 32×5 audio-feature vector is ultimately obtained.

FFT: FFT is used to transfer the input audio data into the frequency domain, and the FFT data are then turned into the natural logarithm of the magnitudes. The generated data are then provided after being downsampled. The result is a 256-element one-dimensional audio feature vector.

Mel-Spec: Short-term fourier transform, which is a series of fourier transforms of a windowed signal moving with a specific hop, is used to transform the input audio data into a frequency domain spectrogram. The Mel scale is then used to process the frequencies. We selected a hop size of 10 ms, a window size of 25 ms, and 80 mel-frequency components for the hyperparameter configuration. After that, a network of two 2D convolutional layers receives the spectrogram data. A $32 \times 40 \times 1$ audio-feature vector is eventually obtained.

4.1.1. Network architecture

The model of Blind DL AI consists of a fully connected three-layered network, a gated recurrent unit [38], and the selected audio encoder previously mentioned. The network's input is the output of the audio encoder. The two hidden levels have 256 nodes each. There are 40 nodes in the output layer, each of which represents one action. The action list was taken from [39]. With respect to the actions, there are two throws on the ground, twelve ground attacks, three ground skills, seven ground movements, two ground guards, twelve air attacks, and two air skills.

4.2. Results

This research used similar environments as in the 2022 DareFightingICE Competition. More specifically, six computers were used that have the same specification, i.e., CPU: Intel(R) Xeon(R) W-2135 CPU@ 3.70 GHz 3.70 GHz, RAM: 16 GB, GPU: NVIDIA Quadro P1000 4 GB VRAM, and OS: Windows 10.

To test our adaptive BGM we used the submissions to the 2022 DareFightingICE Competition Sound Design Track and trained Blind DL AI as mentioned above. The adaptive version of the default and runner-up sound design has its BGM replaced by our adaptive BGM while the rest of the sound effects remain unchanged. We trained Blind DL AI for 900 rounds on DareFightingICE (1 game has 3 rounds) against MCTSAI23i, a weakened version of a sample Monte-Carlo tree search AI [40]. We trained Blind DL AI against MCTSAI23i for 900 rounds for each sound design. Since Blind DL AI has three different audio encoders as an option, the experiments were run on each encoder.

The performance of each trained Blind DL AI was evaluated by making it fight against the aforementioned opponent AI (MCTSAI23i) agent for 90 rounds. The ratio of the number of wins² over 90 rounds ($n = 90$), Eq. (1), and the average HP difference at the end of round r between the trained AI agent and its opponent, Eq. (2), is then

² In the game, the round winner is either the one with a non-zero HP while its opponent's HP has reached zero or the one with the higher HP when the round-length limit of 60 s has reached.

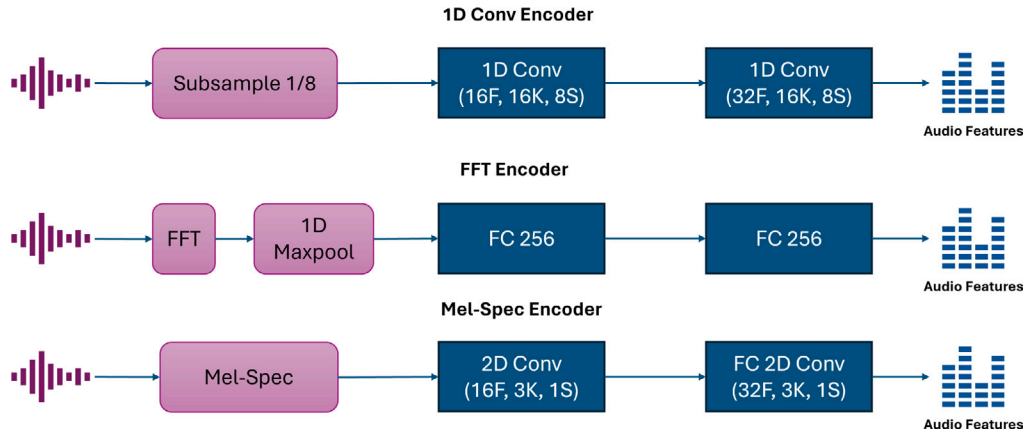


Fig. 4. Audio encoders: 1D-CNN (top), FFT (middle), and Mel-spectrogram (bottom).

Table 1

Blind DL AI performance with default sound design of the 2022 competition and its adaptive version in DareFightingICE.

Sound design	Encoder	win_{ratio}	$avgHP_{diff}$
Default	1D-CNN	0.53	19.92
Default	FFT	0.37	-30.08
Default	Mel-spectrogram	0.54	28.18
Default adaptive	1D-CNN	0.71	59.99
Default adaptive	FFT	0.72	56.09
Default adaptive	Mel-spectrogram	0.61	50.6

Table 2

Blind DL AI performance with runner-up sound design of the 2022 competition and its adaptive version in DareFightingICE.

Sound design	Encoder	win_{ratio}	$avgHP_{diff}$
Runner-up	1D-CNN	0.56	31.68
Runner-up	FFT	0.64	47.73
Runner-up	Mel-spectrogram	0.60	35.02
Runner adaptive	1D-CNN	0.79	75.21
Runner adaptive	FFT	0.66	49
Runner adaptive	Mel-spectrogram	0.62	36.24

Table 3

Blind DL AI performance with winner sound design of the 2022 competition and its adaptive version in DareFightingICE.

Sound design	Encoder	win_{ratio}	$avgHP_{diff}$
Winner	1D-CNN	0.54	12.66
Winner	FFT	0.52	27.03
Winner	Mel-spectrogram	0.56	34.23
Winner adaptive	1D-CNN	0.81	94.78
Winner adaptive	FFT	0.77	88.16
Winner adaptive	Mel-spectrogram	0.7	64.59

calculated. The equations and details above are taken from previous work for the sake of self-contained representation [8].

$$win_{ratio} = \frac{winning\ rounds}{n} \quad (1)$$

$$avgHP_{diff} = \frac{\sum_{r=1}^{r=n} (HP_r^{self} - HP_r^{opp})}{n} \quad (2)$$

As shown from the results in Tables 1, 2, and 3, Blind DL AI performs better on the adaptive version of each sound design in all encoders. This proves the hypothesis that the adaptive BGM was giving useful information. The best AI agent in all sound designs was the blind DL AI with ID-CNN encoder trained on the winner sound design because it had the highest win ratio and HP difference.

Table 4

Blind DL AI performance with the individual groups of instruments of adaptive BGM with winner sound design of the 2022 competition.

Sound design	Encoder	win_{ratio}	$avgHP_{diff}$
Adaptive drums only	1D-CNN	0.64	33.78
Adaptive drums only	FFT	0.69	40.29
Adaptive drums only	Mel-spectrogram	0.62	43.19
Adaptive strings only	1D-CNN	0.61	10.14
Adaptive strings only	FFT	0.64	59.11
Adaptive strings only	Mel-spectrogram	0.60	35.21
Adaptive others only	1D-CNN	0.57	25.33
Adaptive others only	FFT	0.56	3.84
Adaptive others only	Mel-spectrogram	0.57	22.19

4.2.1. Ablation study on individual instrument groups

To investigate the individual impact of each adaptive groups of instruments, we conducted additional experiments where only one groups of instruments (drums, strings, or others) was adaptively modulated at a time while keeping the rest static. The training and evaluation protocol was identical to the primary evaluation. We conducted this experiment with only the winner sound design.

Results from Table 4 show that adaptive drums had the greatest impact on AI performance, with a noticeable improvement in win ratio and average HP difference compared to fully static BGM. Adaptive strings provided moderate improvements, while the adaptive ‘others’ group showed the smallest performance gain. It is also worth noting that while all instrument groups improved performance compared to static BGM, the full adaptive BGM achieved the best overall results.

4.3. The behavior of Blind DL AI agents

This section describes the behavior.³ of three different Blind DL AIs for every sound design. Only the behavior of Blind DL AIs trained on the adaptive BGM was described. For each sound design, the encoder with the best result with respect to the win ratio was chosen. The summary of different behaviors is shown in Table 5

4.3.1. Default sound design

For the default sound design, the best encoder was FFT. This Blind DL AI was the weakest among the selected AIs, and its playing style was moderate. The most used attacks by this AI agent were light kicks. It did not try to avoid the fireball attack. The uniqueness of this AI was that it performed the throw attack which lifts the opponent, MCTSAI23i, in the air.

³ Link to the videos of Blind DL AIs (Player 1: Blind DL AI, Player 2: MCTSAI23i): <http://tinyurl.com/Blind-AI-Vids>.

Table 5
Difference between the different blind DL AI agents.

Sound design	Playing style	Favorite move(s)	Avoid projectiles	Lighting uppercut
Default	Moderate	Light kicks	No	Not used
Runner-up	Defensive	Uppercut and crouch kick	No	Used
Winner	Aggressive	Crouch kicks	No	Used

4.3.2. Runner-up sound design

For the runner-up sound design, the best encoder was 1D-CNN. Its behavior was similar to the default sound design AI agent. The difference was that this AI agent attacked slowly as compared to the other Blind DL AI agents and it played defensively. The most used attack combination of this AI was an uppercut followed by a crouch kick. This AI agent tried to defend itself from MCTSAI23i's jump attacks with uppercuts. It also did the lighting uppercut move. It did not try to avoid the fireball attack.

4.3.3. Winner sound design

For the winner sound design, the best encoder was 1D-CNN. This Blind DL AI was the best among the selected AIs and its playing style was aggressive compared to the other selected AIs. Its most used attack was the crouch kick. This AI agent tried to defend itself from MCTSAI23i's jump attacks with either lighting uppercuts or jumping attacks. It used the lighting uppercut attack more than the runner-up AI agent. It also did not try to avoid the fireball attack.

5. Subjective evaluation

For subjective evaluation we performed two user studies. In one study, the participants just watched two human players fight against each other, which is called the perception study and in the other, the participants played DareFightingICE,⁴ called the playing study. The details of both studies are given below.

In both user studies, statistical tests were conducted to evaluate the significance of the findings. First, we applied the Shapiro–Wilk test to assess the normality of data distributions for both static and adaptive BGM responses, as it is widely recognized for its effectiveness in evaluating normality assumptions in statistical analyses [41]. If the data followed a normal distribution, we conducted a two-tailed t-test; otherwise, we employed a two-tailed Wilcoxon Signed-Rank test, a non-parametric alternative suitable for comparing paired samples without assuming normality [42]. For game performance, however, we performed a one-tailed test based on our hypothesis that adaptive BGM, by providing additional information, would enhance human players' performance. This approach aligns with standard practices in hypothesis testing, where one-tailed tests are used to detect effects in a specific direction.

Prior to both user studies, participants underwent training on the DareFightingICE platform, and all had a background in playing video games. This preparation rendered them well-versed in the platform's mechanics, establishing them as domain experts for DareFightingICE. Although the sample size in our user studies is modest, the training and participants' familiarity with the platform significantly enhanced the quality of insights obtained, ensuring robust and informed feedback.

⁴ Link to the game files used in subjective evaluation: <https://tinyurl.com/A-BGM-Files>.

Table 6
Adaptation recognition questions in the perception study.

Does the background music respond to changes in player 2's HP?
Does the background music reflect changes in the HP of player 1?
Is the background music influenced by the distance between the players?

Table 7
Aesthetic evaluation questions.

I enjoy the background music in the game.
I feel the game's audio (e.g., sound effects, music) enhances my gaming experience.
I think the game's audio fits the mood or style of the game.

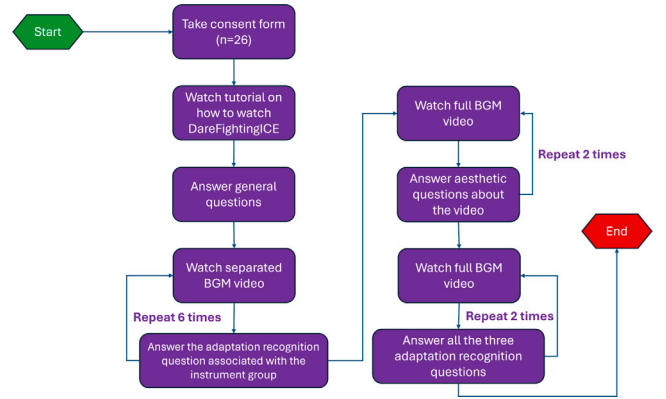


Fig. 5. Protocol of the perception study.

5.1. Perception study

This user study was conducted to find if human players can differentiate between the adaptive BGM and static BGM and if they can correctly differentiate between each group of instruments' connection to the game element. For the questionnaire, we introduced our own questions for the adaptive BGM recognition, and three most related questions from the audio aesthetics factor of GUESS [43] — the Game User Experience Satisfaction Scale — which is widely used in game-related research for evaluating games in user studies. The participants responded to each questionnaire item using a 7-point Likert scale (1 = Strongly Disagree, 4 = Neutral, 7 = Strongly Agree). Questions for both parts are shown in Tables 6 and 7. The results of this user study are presented later in this section.

Out of 26 participants, 8% were extremely familiar with fighting games before the user study, 11% were very familiar, 50% were familiar, 27% were slightly familiar, and 4% were not familiar at all.

The user study was divided into two parts. In the first part, the participants watched six videos, two for each of the three groups of instruments, where one video had the assigned group of instruments adaptively played and the other was static. This was done to later introduce the participants to the full adaptive BGM (played by all groups of instruments) and to see if they could correctly differentiate between the adaptive BGM's groups of instruments separately. In the second part, the participants first watched two videos — one with full adaptive BGM and the other with full static — and answered the aforementioned aesthetic questions. After the aesthetic part, they again watched one adaptive video and one static video and answered the adaptation recognition questions. The user study protocol is visualized in Fig. 5; in the protocol, the target videos were randomly shuffled and individually shown at each iteration together with the relevant question(s).

5.1.1. Results

This user study had two factors, aesthetic evaluation and adaptation recognition. Aesthetic evaluation in this research evaluates both static

Table 8

Aesthetic evaluation (Full BGM) results.

BGM	Mean	Std
Static BGM	15.19	3.75
Adaptive BGM	16.92**	2.44

** p -value < 0.01.**Table 9**

Adaptation recognition (Separated) results.

BGM	Mean	Std
Static BGM	10.52	3.75
Adaptive BGM	14.15***	2.44

*** p -value < 0.001.**Table 10**

Adaptation recognition (Full BGM) results.

BGM	Mean	Std
Static BGM	10.12	3.10
Adaptive BGM	13.62***	3.07

*** p -value < 0.001.

and adaptive BGMS on music aesthetics. Adaptation recognition was defined as the ability of participants to correctly recognize the adaptive elements of the adaptive BGM as well as recognize which BGM was adaptive and which was static.

For the aesthetic evaluation, the results in Table 8 showed that there is a statistically significant difference between the adaptive BGM and the static BGM. This indicates that, even after modifying the BGM, it not only retained its aesthetic value but also improved it, which is a positive outcome.

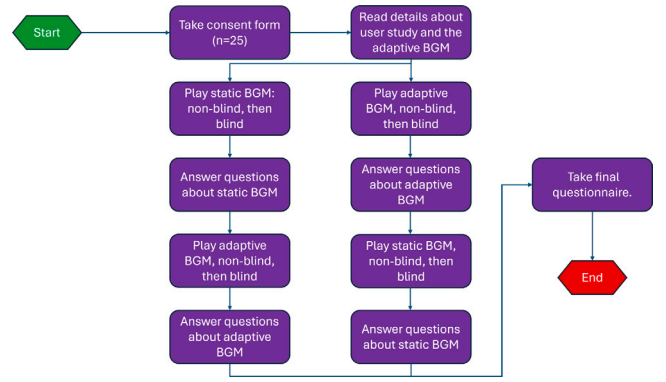
The adaptation recognition was divided into two parts as previously mentioned. Both results from separate and full BGMS showed that with statistical significance, the participants could more correctly tell which game element a group of instruments was connected to (Table 9), and the adaptive BGM was more recognizable (Table 10).

This user study showed that human players can perceive helpful information on game elements from the adaptive BGM, which shows its effectiveness.

5.2. Playing study

The second user study was conducted to see if human players would perform better with the adaptive BGM as compared to the static BGM and to evaluate the adaptive BGM on five different factors of the video game satisfaction scale. These factors were taken from GUESS and were audio aesthetics, usability/playability, play engrossment, enjoyment, and personal gratification. We used only these five because they were the most relevant factors when it comes to our research and we used slightly modified three most related questions for our research from each factor. Same as the perception study, the participants responded to each questionnaire item using a 7-point Likert scale. A total of 27 participants were recruited from Ritsumeikan University. All of the participants had prior experience with the DareFightingICE platform. Two participants were removed from the analysis since their data were deemed outliers due to their low win rate for static BGM against the opponent AI agent, with one of them getting 0%. Of the remaining 25 participants, 8% were extremely familiar with fighting games before the user study, 16% were very familiar, 44% were familiar, 28% were slightly familiar, 4% were not familiar at all.

This user study was also divided into two parts. We wanted to evaluate our adaptive BGM on both players with and without vision, but we used two game modes due to the unavailability of visually impaired players at the time of user study. These two game modes were blind and non-blind; in the blind mode, the users would only see a black

**Fig. 6.** Protocol of the playing study.**Table 11**

Adaptation recognition questions in the playing study.

Were you able to tell the difference between adaptive and static BGM? Please give a brief explanation on how?
Please write the pros and cons of adaptive BGM in your opinion.
In your opinion, what can be done to improve the adaptive BGM?

screen while playing the game without being blindfolded. The reason for not blindfolding the players was to make sure they focus on the sound and the gameplay without having to worry about the location of controls as VIPs are already accustomed to the controls. This approach of using sighted players as a proxy for VIPs while making them blind has been done in the past when VIPs were not available [44,45] or were very few [46]. While using players with vision as a proxy for VIPs was not in any way a perfect solution, it had been observed that VIPs tend to perform better than blindfolded or blinded-sighted players [46,47]. As a result, this approach provided a baseline for the expected performance from VIPs on the adaptive BGM.

The participants played a total of 12 games, three games for each combination of blind and non-blind modes for both adaptive and static BGMS against the above-mentioned MCTSAI23i. The participants were divided into two groups, one playing the adaptive BGM first and the other static. After they played both blind and non-blind for one BGM they were asked to answer a questionnaire containing questions about the aforementioned five factors of GUESS, with three questions for each factor. At the end of the user study, the participants were told to answer a final questionnaire asking for their opinion about adaptive BGM. The questions used in this questionnaire are shown in Table 11, and the protocol for this user study is shown in Fig. 6.

Apart from answering the questionnaires, the performance of the participants in both blind and non-blind modes for both of the BGMS was recorded for analysis. To evaluate the performance of participants in both blind and non-blind modes we used the approach mentioned in previous work [8], which was also how the results from the sound design track of the DareFightingICE Competition were calculated. However, the evaluation process has been improved since then by adding the win ratio in the Eq. (4) to calculate the normalized performance, where the win ratio is defined as the ratio of rounds won by the player out of the total rounds. This was done to improve the existing evaluation metric using only the HP difference to measure the performance, which does not represent the player's overall performance.

We took the evaluation process of the 2024 DareFightingICE Sound Design Competition where they defined γ (4) as the new normalized performance by multiplying δ (5), the old normalized performance, by the win ratio, where due to the initial setting of HP , min and max are -400 and 400 , respectively. In the end, They used the "Performance Retention Rate (PRR)" (3) from previous work [8] to show how much performance was retained from non-blind play to blind play. This PRR

Table 12
Results from factor: Usability/Playability.

BGM	Mean	Std
Static BGM	14.96	3.41
Adaptive BGM	14.92	3.66

Table 13
Results from factor: Play engrossment.

BGM	Mean	Std
Static BGM	13.16	3.36
Adaptive BGM	13.20	3.66

Table 14
Results from factor: Enjoyment.

BGM	Mean	Std
Static BGM	15.04	3.21
Adaptive BGM	15.72*	3.37

* p -value < 0.05.

Table 15
Results from factor: Personal gratification.

BGM	Mean	Std
Static BGM	17.24	1.96
Adaptive BGM	16.80	2.12

Table 16
Results from factor: Audio aesthetics.

BGM	Mean	Std
Static BGM	15.60	2.35
Adaptive BGM	16.92**	2.34

** p -value < 0.01.

was also used in the sound design track of the 2022 DareFightingICE Competition.

$$PRR = \frac{\text{average}(\gamma_{blind})}{\text{average}(\gamma_{non-blind})} \quad (3)$$

$$\gamma = \text{winratio} \times \delta \quad (4)$$

$$\delta = \frac{d - \min}{\max - \min} \quad (5)$$

$$d = \text{HP}_{Player} - \text{HP}_{AI} \quad (6)$$

5.2.1. Results

We hypothesized that there would be little to no difference between adaptive BGM and static BGM when it comes to the five GUESS factors in use as the overall game remained the same with just BGM being changed.

The results from Table 12, 13, 14, and 15 showed that there was no significant difference between the two types of BGM when it comes to usability/playability, play engrossment, and personal gratification. For audio aesthetics, however, the results from Table 16 showed that there was a significant difference between them, with adaptive BGM scoring higher. These results demonstrated that our adaptive BGM not only improved the aesthetic quality compared to the static version, consistent with findings from the perception study, but also had no negative impact on other factors. Additionally, there was a significant difference in favor of adaptive BGM on the enjoyment factor, correlating with the audio aesthetic results, as more enjoyable music leads to a more enjoyable gaming experience.

The results from Table 17 showed that the participants performed better with adaptive BGM as compared to static BGM. The PRR showed that the participants were able to retain a lot more of their performance

Table 17
PRR of adaptive and static BGM.

PRR	
Adaptive BGM	Static BGM
0.84*	0.65

* p -value < 0.05.

Table 18
Sentiment analysis of the open-ended questions.

Question	Sentiment
1	Generally positive
2	Mixed, Slightly positive
3	Mixed, Slightly positive

from non-blind play to blind play compared to static BGM. This added to the proof of our previously mentioned hypothesis that the adaptive BGM was indeed giving useful information for both players with and without vision.

Finally, for the evaluation and analysis of the open-ended questions asked at the end of the playing study, we utilized ChatGPT⁵ to summarize some key findings. These findings were then double-checked by the first two authors to ensure their accuracy. We used ChatGPT (GPT-4o) inspired by existing studies [48,49], it was shown to be adequate when it comes to natural language processing tasks. The findings of our analysis are shown below.

For the sentiment analysis, the sentiment in responses to the questions shown in Table 11 was generally positive, with participants describing the advantages of adaptive BGM, such as better ambiance, enhanced immersion, and information retrieval. However, there were also mixed sentiments, especially regarding potential challenges like a steep learning curve regarding sound recognition. Constructive suggestions for improvement indicated an overall positive outlook on the potential enhancements to adaptive BGM. The information regarding the sentiment analysis and the positive and negative elements of adaptive BGM are shown in Tables 18 and 19.

For the first question, most of the participants reported having noticed the difference between static and adaptive BGM, with some of them reporting that the adaptive BGM increased their immersion. However, while participants qualitatively reported increased immersion, our quantitative measures did not find significant differences in play engrossment, suggesting a need for future studies using validated immersion scales to clarify these findings.

“Yes. In the blind static mode, I had absolutely no idea whether I had an edge on my opponent or not. Also, I believe it somehow enhances the immersion in the game (when the volume goes up, tension goes up as well).” Participant5

For the second question, the advantages that were most commonly described by the participants were information enhancement, and engagement and immersion. They appreciated that adaptive BGM provided more information about the game and mentioned that adaptive BGM made the game more engaging and immersive. This can be seen by the results above by their performance and from adaptive BGM being aesthetically better than static. The most commonly described disadvantages were learning curve, and noise or confusion. The participants described a certain degree of difficulty in recognizing the adaptive BGM and mentioned that it sometimes confused them due to too much information being given.

“Pros: I feel more immerse to the game. The situation on the game is easy to understand. Cons: When a game has a lot of sounds, it makes it more confusing. Especially if we are not familiar with the game” Participant15

For the last question, the most common improvements the participants mentioned were better volume management and clarity in sound

⁵ <https://chat.openai.com/>.

Table 19
Positive and negative elements of adaptive BGM described in open-ended questions.

Question	Positive element(s)	Negative element(s)
1	Ambiance, Enhanced immersion, Information retrieval	None
2	Information enhancement, Engagement	Confusion and distracting noise
3	Feedback on enhancing volume management	Steep learning curve

differentiation. It was observed that participants at times found it hard to distinguish between the different groups of instruments.

“Make the volume more constant (or put an instrument in every BGM element) so that we can know at any moment the changes in the game situation. (sometimes it is hard to perceive the volume change)” Participant5

6. Discussion

The players perform better on adaptive BGM, especially in blind mode because our adaptive BGM gives information about the closeness of the players and the HP of both players. It is, however, hard for some players to grasp the changes in BGM, especially for the group of instruments connected to HP of Player 2; this is a result of us choosing the least recognizable group for HP of Player 2. Moreover, the players are able to distinguish the changes in the drums as it is the most prominent and consistent part of the BGM.

To address challenges in distinguishing instrument groups (e.g., the opponent HP-linked ‘Others’), future implementations could leverage spatial audio panning to emphasize distinct game elements. For instance, opponent HP cues could be panned to the right channel for Player 2 and to the left for Player 1’s HP. Such an approach may enhance clarity without overwhelming listeners. Future work could also explore the use of additional or alternative instruments to ensure that players can easily perceive the information without experiencing auditory overload.

While our study did not find a statistically significant difference in ‘play engrossment’ between adaptive and static BGM, participants reported heightened enjoyment and perceived audio aesthetics with adaptive BGM. These factors, while distinct from engrossment, may contribute to a subjective sense of immersion by enhancing emotional engagement and situational awareness. Future work could incorporate validated immersion-specific metrics to disentangle these effects.

We selected the levels for the volume for the groups of instruments through testing conducted before the evaluation to find the best combination due to the lack of research on this subject. We found the approach of increasing the volume as the value of the elements decreased to be more effective than the opposite, since it gives a sense of urgency as the round progressed and it was found in testing that volume increasing is easier to detect and distinguish than volume decreasing. The specific levels for volume are just implemented to give a gradual sense of increase and to make sure the BGM does not become too loud when the HP or PD is the lowest.

In terms of the generalizability of our approach, we argue that our approach and findings can help not only fighting games but other genres of video games as well. Since we show how an adaptive BGM can help both players with and without vision in fighting games, the same can be said for other genres like role-playing games for example where music gets louder the closer you are to an enemy or to your objective. Although we cannot be certain that the level of impact adaptive BGM have on DareFightingICE to be similar when it comes to other genres, we believe with some modification adaptive BGM can be implemented for a target game.

This approach provided a baseline performance that we can expect from VIPs using adaptive BGM in DareFightingICE. While not including VIPs in the evaluation of adaptive BGM is a concern, the performance of blinded-sighted players shows that VIPs will perform equally or better as they are more accustomed to playing with only sound.

7. Limitations

Both objective and subjective evaluation are done against a weak opponent which can affect the results, and there exists some room for future work to try this approach against a variety of opponent AI agents. When it comes to the subjective evaluation, as mentioned previously the paper did not include VIPs in the evaluation process. While blinding sighted players can give a baseline to the performance of VIPs, VIPs are still needed in the final evaluation process before the system’s deployment. There is still a need for VIPs in the evaluation process. Furthermore, as mentioned by some of the participants the adaptive BGM can be a bit noisy, which makes it harder for people to distinguish between different groups of instruments. This can be a shortcoming of our approach that only modifies the volume of the instrument groups. A better approach for the future would be to try modifying other sound elements such as pitch or tempo. This approach would make it easier to distinguish between different groups. However, it would be a challenge to come up with an appropriate range for an element like the pitch of the BGM as pitch can be more sensitive to change [50] and can change the aesthetic of the BGM drastically [51].

While volume modulation risks causing auditory overload, integrating tempo or pitch adjustments could offer more nuanced differentiation. Pitch modulation may significantly improve distinguishability, but can lead to undesirable harmonic dissonance if not carefully managed. Tempo changes, by contrast, could convey game urgency but risk disrupting rhythmic consistency. For example, gradual tempo increases during close combat might reflect rising intensity without compromising musical harmony. Pitch shifts within a musical key (e.g., ± 2 semitones) could signal HP thresholds while preserving aesthetic coherence. However, excessive pitch changes may alter the BGM’s emotional tone, requiring careful calibration. Hybrid approaches such as combining volume, tempo, and subtle pitch modulation could strike a balance between informativeness and musical integrity, though they demand thoughtful design and empirical validation to prevent sensory overload.

8. Conclusions

This paper presented a rule-based adaptive background music (BGM) system that consists of three different categories of instruments playing the BGM of the winner sound design from the 2022 DareFightingICE Competition. The proposed adaptive BGM adapts by changing the volumes of the mentioned different categories of instruments. Each category of instruments was connected to a different element of the game. This paper also showed that the performance of a deep reinforcement learning AI agent using only audio data as its input, called Blind DL AI, improved while playing with the adaptive BGM as compared to playing with the original (non-adaptive) BGM, called static BGM. The ablation study confirmed that different groups of instruments contribute differently to gameplay performance, highlighting the particular effectiveness of drums as an adaptive indicator in fighting game contexts. Lastly, this paper also introduced two user studies, the results of which showed that human players can recognize the adaptation of the adaptive BGM and that human players also perform better with adaptive BGM just like Blind DL AI. This meant that, as with Blind DL AI, human players could get useful information from the adaptive BGM. The study also showed that there were no differences in usability/playability, play engrossment, and personal gratification between adaptive and static BGM but adaptive BGM scored better on audio aesthetics, and enjoyment.

In the future, we are planning to use a deep learning approach to the adaptation of BGM. This is because the adaptive BGM using deep reinforcement learning would be more informative than the one resulting from the rule-based one proposed in this work, as we plan to use Blind DL AI's performance as part of the reward function. As for only using volume in the adaptation process, we plan to use other elements in the adaptation of the BGM as well, such as pitch and tempo.

CRedit authorship contribution statement

Ibrahim Khan: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Thai Van Nguyen:** Writing – review & editing, Supervision, Software. **Ruck Thawonmas:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors of this paper would like to express their gratitude to Chuang Boyu, Shouchen Ye, Tian Yuchen, and Jun Tanabe from the Intelligent Computer Entertainment laboratory, Ritsumeikan University for their help in translating the user study details into Japanese for Japanese-speaking participants. This work was supported in part by Grant-in-Aid for Scientific Research (C), No. 25K15412.

Data availability

The code and relevant information regarding the data is given in the paper.

References

- [1] G.N. Yannakakis, J. Togelius, *Artificial Intelligence and Games*, Springer, New York, 2018, <http://dx.doi.org/10.1007/978-3-319-63519-4>.
- [2] S. Risi, M. Preuss, From chess and atari to StarCraft and beyond: How game AI is driving the world of AI, *Künstl. Intell.* 34 (2020) 7–17, <http://dx.doi.org/10.1007/s13218-020-00647-w>.
- [3] S. Gormanley, Audio immersion in games — a case study using an online game with background music and sound effects, *Comput. Games J.* (2013) 103–124, <http://dx.doi.org/10.1007/BF03392344>.
- [4] I. Ekman, On the desire to not kill your players: Rethinking sound in pervasive and mixed reality games, in: *FDG*, 2013, pp. 142–149.
- [5] Capcom, *Street Fighter*, Capcom, Osaka, Japan, 1987.
- [6] Bandai Namco Entertainment, *Tekken*, Bandai Namco Entertainment, Minato, Tokyo, 1994.
- [7] Midway Games, *Mortal Kombat*, Midway Games, Chicago, Illinois, U.S., 1992.
- [8] I. Khan, T.V. Nguyen, X. Dai, R. Thawonmas, DareFightingICE competition: A fighting game sound design and AI competition, in: 2022 IEEE Conference on Games, CoG, Beijing, China, 2022, pp. 478–485, <http://dx.doi.org/10.1109/CoG51982.2022.9893624>.
- [9] F. Lu, K. Yamamoto, L.H. Nomura, S. Mizuno, Y. Lee, R. Thawonmas, Fighting game artificial intelligence competition platform, in: 2013 IEEE 2nd Global Conference on Consumer Electronics, GCCE, Tokyo, Japan, 2013, pp. 320–323, <http://dx.doi.org/10.1109/GCCE.2013.6664844>.
- [10] I. Khan, T. Van Nguyen, C. Nimpattavong, R. Thawonmas, Fighting game adaptive background music for improved gameplay, in: 2023 IEEE Conference on Games, CoG, Boston, MA, USA, 2023, pp. 1–2, <http://dx.doi.org/10.1109/CoG57401.2023.10333245>.
- [11] R. Andrade, M.J. Rogerson, J. Waycott, S. Baker, F. Vetere, Playing blind: Revealing the world of gamers with visual impairment, in: Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems, 2019, pp. 1–14, <http://dx.doi.org/10.1145/3290605.3300346>.
- [12] J. Broderick, J. Duggan, S. Redfern, The importance of spatial audio in modern games and virtual environments, in: 2018 IEEE Games, Entertainment, Media Conference, GEM, Galway, Ireland, 2018, pp. 1–9, <http://dx.doi.org/10.1109/GEM.2018.8516445>.
- [13] Luise Haehn, Sabine J. Schlittmeier, Christian Böffel, Exploring the impact of ambient and character sounds on player experience in video games, *Appl. Sci.* 14 (2) (2024) 583, <http://dx.doi.org/10.3390/app14020583>.
- [14] K. Tuuri, O. Koskela, J. Vahlo, H. Tissari, Identifying the impact of game music both within and beyond gameplay, in: J.C.S. Baalsrud Hauge, J. Cardoso, L. Roque, P.A. Gonzalez-Calero (Eds.), *Entertainment Computing – ICEC 2021, ICEC 2021*, in: Lecture Notes in Computer Science, vol. 13056, Springer, Cham, http://dx.doi.org/10.1007/978-3-030-89394-1_33.
- [15] JaeHwan Byun, Christian S. Loh, Audial engagement: Effects of game sound on learner engagement in digital game-based learning environments, *Comput. Hum. Behav.* 46 (2015) 129–138, <http://dx.doi.org/10.1016/j.chb.2014.12.052>.
- [16] J. Zhang, X. Fu, The influence of background music of video games on immersion, *J. Psychol. Psychother.* 5 (4) (2015) 1.
- [17] Felix Andersen, Calvin Leonard King, Alexander A.S. Gunawan, Audio influence on game atmosphere during various game events, *Procedia Comput. Sci.* 179 (2021) 222–231, <http://dx.doi.org/10.1016/j.procs.2021.01.001>.
- [18] Colby Johanson, Regan L. Mandryk, Scaffolding player location awareness through audio cues in first-person shooters, in: Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems, CHI '16, Association for Computing Machinery, New York, NY, USA, 2016, pp. 3450–3461, <http://dx.doi.org/10.1145/2858036.2858172>.
- [19] Patrick Ng, Keith Nesbitt, Informative sound design in video games, in: Proceedings of the 9th Australasian Conference on Interactive Entertainment: Matters of Life and Death, IE '13, Association for Computing Machinery, New York, NY, USA, 2013, p. 9, <http://dx.doi.org/10.1145/2513002.2513015>, 1–9.
- [20] C. Chai, B.T. Lau, Z. Pan, Hungry Cat—a serious game for conveying spatial information to the visually impaired, *Multimodal Technol. Interact.* 3 (1) (2019) 12, <http://dx.doi.org/10.3390/mti3010012>.
- [21] V. Nair, J.L. Karp, S. Silverman, M. Kalra, H. Lehv, F. Jamil, B.A. Smith, Navstick: Making video games blind-accessible via the ability to look around, in: The 34th Annual ACM Symposium on User Interface Software and Technology, 2021, pp. 538–551, <http://dx.doi.org/10.1145/3472749.3474768>.
- [22] Bandai Namco Entertainment, *Tekken 7*, Bandai Namco Entertainment, Minato, Tokyo, 2015.
- [23] NetherRealm Studios, *Mortal Kombat 11*, NetherRealm Studios, Chicago, Illinois, US, 2019.
- [24] Double Helix Games, *Killer Instinct*, Double Helix Games, Irvine, California, US, 2013.
- [25] Capcom, *Street Fighter 6*, Capcom, Chūō-ku, Osaka, Japan, 2023.
- [26] K.B. McAlpine, M. Bett, J. Scanlan, Approaches to creating real-time adaptive music in interactive entertainment: a musical perspective, in: 35th Audio Engineering Society International Conference 2009: Audio for Games: Proceedings Article 14 Audio Engineering Society, 2009, <https://secure.aes.org/forum/pubs/conferences/?elib=15168>.
- [27] Mirjam Eladhari, Rik Nieuwdorp, Mikael Fridenfalk, The soundtrack of your mind: mind music - adaptive audio for game characters, in: Proceedings of the 2006 ACM SIGCHI International Conference on Advances in Computer Entertainment Technology, ACE '06, Association for Computing Machinery, New York, NY, USA, 2006, pp. 54–es, <http://dx.doi.org/10.1145/1178823.1178887>.
- [28] M.López. Ibáñez, N. Álvarez, F. Peinado, Towards an emotion-driven adaptive system for video game music, in: A. Cheok, M. Inami, T. Romão (Eds.), *Advances in Computer Entertainment Technology, ACE 2017*, in: Lecture Notes in Computer Science, vol. 10714, Springer, Cham, 2018, http://dx.doi.org/10.1007/978-3-319-76270-8_25.
- [29] P.E. Hutchings, J. McCormack, Adaptive music composition for games, *IEEE Trans. Games* 12 (3) (2020) 270–280, <http://dx.doi.org/10.1109/TG.2019.2921979>.
- [30] C. Plut, P. Pasquier, Music matters: An empirical study on the effects of adaptive music on experienced and perceived player affect, in: 2019 IEEE Conference on Games, CoG, London, UK, 2019, pp. 1–8, <http://dx.doi.org/10.1109/CIG.2019.8847951>.
- [31] M. Makhmutov, J.A. Brown, M. Surkov, A. Timchenko, K. Timchenko, Adaptive game soundtrack tempo based on players' actions, in: 2022 IEEE Conference on Games, CoG, Beijing, China, 2022, pp. 441–448, <http://dx.doi.org/10.1109/CoG51982.2022.9893604>.
- [32] Mane6, *Them's Fightin' Herds*, Mane6, US, 2020.
- [33] C. Nimpattavong, I. Khan, T. Van Nguyen, R. Thawonmas, W. Choensawat, K. Sookhanaphibarn, Improving data transfer efficiency for AIs in the Dare-FightingICE using gRPC, in: 2023 8th International Conference on Business and Industrial Research, ICBIR, Bangkok, Thailand, 2023, pp. 286–290, <http://dx.doi.org/10.1109/ICBIR57571.2023.10147629>.
- [34] Moises systems, 2024, <https://moises.ai/>, (Accessed 17 July 2025).
- [35] Carl B. Hancock, Aesthetic responses of music and non-music majors to gradual pitch center changes, *Bull. Counc. Res. Music. Educ.* 1 (178) (2008) 85–94, <http://dx.doi.org/10.2307/40319341>.

- [36] V. Nair, S.E. Ma, R.E. Gonzalez Penuela, Y. He, K. Lin, M. Hayes, . . . , B.A. Smith, Uncovering visually impaired gamers' preferences for spatial awareness tools within video games, in: *Proceedings of the 24th International ACM SIGACCESS Conference on Computers and Accessibility*, 2022, pp. 1–16.
- [37] T. Van Nguyen, X. Dai, I. Khan, R. Thawonmas, H.V. Pham, A deep reinforcement learning blind AI in DareFightingICE, in: *2022 IEEE Conference on Games, CoG*, Beijing, China, 2022, pp. 632–637, <http://dx.doi.org/10.1109/CoG51982.2022.9893718>.
- [38] K. Cho, B. Van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, Y. Bengio, Learning phrase representations using rnn encoder–decoder for statistical machine translation, in: *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing, EMNLP*, 2014, pp. 1724–1734, <http://dx.doi.org/10.48550/arXiv.1406.1078>.
- [39] R. Liang, Y. Zhu, Z. Tang, M. Yang, X. Zhu, Proximal policy optimization with elo-based opponent selection and combination with enhanced rolling horizon evolution algorithm, in: *2021 IEEE Conference on Games, CoG*, Copenhagen, Denmark, 2021, pp. 1–4, <http://dx.doi.org/10.1109/CoG52621.2021.9619146>.
- [40] Makoto Ishihara, Taichi Miyazaki, Chun Yin Chu, Tomohiro Harada, Ruck Thawonmas, Applying and improving Monte-Carlo tree search in a fighting game AI, in: *Proceedings of the 13th International Conference on Advances in Computer Entertainment Technology, ACE '16*, Association for Computing Machinery, New York, NY, USA, 2016, p. 27, <http://dx.doi.org/10.1145/3001773.3001797>, 1–6.
- [41] Samuel Sanford Shapiro, Martin B. Wilk, An analysis of variance test for normality (complete samples), *Biom.* 52 (3–4) (1965) 591–611.
- [42] G.E. Noether, Introduction to wilcoxon (1945) individual comparisons by ranking methods, in: S. Kotz, N.L. Johnson (Eds.), *Breakthroughs in Statistics*, in: *Springer Series in Statistics*, Springer, New York, NY, 1992, http://dx.doi.org/10.1007/978-1-4612-4380-9_15.
- [43] M.H. Phan, J.R. Keebler, B.S. Chaparro, The development and validation of the game user experience satisfaction scale (GUESS), *Hum. Factors* 58 (8) (2016) 1217–1247, <http://dx.doi.org/10.1177/0018720816669646>.
- [44] J.F.P. Cheiran, L. Nedel, M.S. Pimenta, Inclusive games: A multimodal experience for blind players, in: *Proceedings of the 2011 Brazilian Symposium on Games and Digital Entertainment*, 2011, pp. 164–172.
- [45] A. Adkins, K. Kohm, R. Zhang, N. Gustafson, Lost in space: An audio maze game for the visually impaired, in: *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*, 2020, pp. 1–6.
- [46] N. Mohd Norowi, H. Azman, N. Wahiza Abdul Wahat, Apple swipe: A mobile game apps for visually impaired users using binaural sounds, in: *Asian CHI Symposium*, 2021, pp. 196–201.
- [47] A.M. Silverman, The perils of playing blind: Problems with blindness simulation and a better way to teach about blindness, *J. Blind. Innov. Res.* 5 (2) (2015).
- [48] Wang Jiaan, Liang Yunlong, Meng Fandong, Sun Zengkui, Shi Haoxiang, Li Zhixu, Xu Jinan, Qu Jianfeng, Zhou Jie, Is ChatGPT a Good NLG Evaluator? A preliminary study, *Proceedings of the 4th New Frontiers in Summarization Workshop* (2023) <http://dx.doi.org/10.18653/v1/2023.newssum-1.1>.
- [49] Z. Wang, Q. Xie, Z. Ding, Y. Feng, R. Xia, Is ChatGPT a good sentiment analyzer? A preliminary study, 2023, arXiv preprint [arXiv:2304.04339](https://arxiv.org/abs/2304.04339).
- [50] Josh H. McDermott, Andrew J. Oxenham, Music perception, pitch, and the auditory system, *Curr. Opin. Neurobiol.* 18 (4) (2008) 452–463, <http://dx.doi.org/10.1016/j.conb.2008.09.005>.
- [51] J. Ley-Flores, E. Alshami, A. Singh, et al., Effects of pitch and musical sounds on body-representations when moving with sound, *Sci. Rep.* 12 (2022) 2676, <http://dx.doi.org/10.1038/s41598-022-06210-x>.